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Ambient-temperature aging-driven oxygen vacancy engineering of α-MnO_2/GO for near-theoretical lithium storage

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ABSTRACT

Oxygen vacancy engineering has emerged as a powerful strategy to enhance the electrochemical performance of transition metal oxides; however, achieving high defect density without compromising structural stability remains a fundamental challenge. Herein, we report an ambient-temperature, aging-driven defect engineering strategy that enables the formation of oxygen-vacancy-rich α-MnO_2 nanoparticles uniformly anchored on graphene, which delivers a near-theoretical lithium storage capacity with superior cycling durability. Prolonged aging induced continuous secondary reactions that promote the structure evolution of oxygen-vacancy-enriched α-MnO_2, as confirmed by EPR and XPS analyses. The resulting defect-rich MnO_2 exhibited enhanced intrinsic electronic conductivity and accelerated lithium-ion kinetics, resulting in a high reversible capacity of 1175 mAh g^{-1} at 0.1 A g^{-1}, approaching the theoretical limit. Furthermore, a mild low-temperature heat treatment (95 °C) successfully activated the MnO_2/graphene interface via Mn-O-C configuration while preserving oxygen vacancies, leading to accelerated interfacial charge transfer and outstanding cycling stability. Notably, the optimized composite exhibited sustained activation behavior, with capacity increasing to 977 mAh g^{-1} over 400 cycles. This work highlights a scalable and energy-efficient route for interfacial defect engineering in metal oxide/graphene systems, providing a general strategy to achieve high defect density, fast kinetics, and structural durability for advanced lithium-ion battery anodes.

1. Introduction

Lithium-ion batteries (LIBs) have become indispensable energy storage systems for portable electronics and electric vehicles, driving continuous demand for electrode materials with higher energy density and long-term stability [1–3]. Among various candidates, manganese dioxide (MnO_2) has attracted considerable attention owing to its high theoretical capacity (~1232 mAh g^{-1}), natural abundance, low cost, and environmental benignity [4–9]. However, the practical application of MnO_2-based anodes remains severely limited by intrinsically low electronic conductivity, sluggish lithium-ion diffusion kinetics, and pronounced structural degradation during repeated lithiation and delithiation [10–12].

To address these limitations, extensive efforts have focused on defect engineering, particularly oxygen vacancy modulation, as an effective strategy for enhancing electronic conductivity and redox activity in transition metal oxides [13–15]. Oxygen vacancies introduce localized electronic states, enhance intrinsic electrical conductivity, and modulate the coordination environment of metal centers, thereby facilitating charge transfer and accelerating surface redox kinetics [16–19]. In MnO_2, oxygen vacancies not only create additional electrochemically active sites but also promote electrolyte-ion adsorption/diffusion and

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facilitate reversible Mn valence transitions, leading to significantly enhanced pseudocapacitive charge storage behavior [20–22]. Nevertheless, achieving an optimal concentration of oxygen vacancies in MnO_2 without sacrificing structural stability remains a fundamental challenge [23–25]. Conventional approaches, including high-temperature thermal reduction, hydrogen annealing, plasma treatment, or chemical reducing agents, often induce phase instability, particle coarsening, or irreversible structural damage, thereby leaving a persistent trade-off between defect density and cycling durability [26–28].

Meanwhile, hybridization of MnO_2 with carbonaceous materials such as graphene or graphene oxide (GO) has been widely explored to enhance electrical conductivity and buffer volume expansion [29–31]. While MnO_2/graphene composites indeed offer improved charge transport and mechanical stability, most reported studies primarily emphasize composite fabrication rather than deliberate control over the defect chemistry of MnO_2 [32–36]. In particular, oxygen vacancy generation and interfacial stabilization are often treated as secondary or uncontrolled outcomes, rather than being co-designed as integral components of electrode architecture [37,38]. This lack of defect–interface co-design limits the full exploitation of MnO_2 toward its theoretical capacity and long-term durability.

From this perspective, the chemical environment of the Hummers' method—originally developed for GO synthesis—offers an underexplored opportunity for defect engineering [39–41]. The strong oxidative conditions, residual oxidants, and prolonged aging process inherent to this method can induce continuous secondary reactions that influence the structural evolution of transition metal oxides [42,43]. Unlike conventional post-synthetic thermal or chemical reduction approaches, the aging stage of the Hummers' method represents a synthesis-integrated strategy for regulating oxygen-vacancy formation in situ during composite growth by controlling the aging time under ambient conditions. Nevertheless, despite its widespread use, the aging stage in the Hummers' process has rarely been examined as an active parameter for tailoring defect chemistry, and its role in regulating the electrochemical properties of metal oxides remains largely unexplored.

Herein, we report an aging-driven synthesis strategy to regulate the defect structure of MnO_2 within MnO_2/graphene composites. By systematically controlling the aging time in a modified Hummers' process, we aim to elucidate how aging affects the nucleation behavior, defect formation, and interfacial characteristics of MnO_2. Furthermore, a subsequent low-temperature treatment is introduced to selectively modulate the MnO_2/graphene interface, enabling decoupled control over defect generation and structural stabilization. This work highlights aging as a practical and energy-efficient strategy for oxygen-vacancy engineering, providing an alternative to conventional thermal and chemical reduction methods while offering new insight into the rational design of metal oxide/carbon composites for advanced energy storage.

2. Experimental

2.1. Synthesis of MnG_xh (x = aging time)

Graphite powder (400 mg, 99%, Sigma-Aldrich) was dispersed in 15 mL of sulfuric acid (H_2SO_4, 98%, DaeJung) and sonicated for 30 min. The resulting suspension was then stirred at 300 rpm in an ice bath, and the temperature was maintained below 10 °C. While keeping the solution cold, 5.0 g of $KMnO_4$ was slowly added over 6 h to ensure homogeneous mixing and sufficient oxidation. The mixture was subsequently aged for specific durations (0, 24, 48, and 72 h) under continuous stirring at 300 rpm. After the aging process, the solution was maintained at 35 °C for 24 h in an oil bath, followed by the addition of 200 mL of DI water. The product was then centrifuged and washed several times to remove impurities and subsequently neutralized by dialysis using a dialysis tube for several days until the pH reached 7. Finally, MnG_xh powders were obtained by freeze-drying at −45 °C for 24 h.

2.2. Synthesis of MnG_72h_95 °C_3h

The synthesis procedure of the MnG_72h_95 °C_3h was similar to that of the MnG_xh samples, except for the aging and heat-treatment conditions. The aging time was fixed at 72 h. After the aging process, the solution was maintained at 95 °C for 3 h in an oil bath. Subsequently, 200 mL of DI water was added, and the product was centrifuged and washed several times with DI water, followed by neutralization. Finally, the MnG_72h_95°C_3h powder was obtained by freeze-drying at −45 °C for 24 h.

2.3. Characterization

The morphologies of the MnG samples and the degree of volume expansion before and after electrochemical testing were investigated using a field-emission scanning electron microscope (FE-SEM, Hitachi, SU8000). For cross-sectional observation, the electrodes were separated using a post-mortem method and fractured under liquid nitrogen. A transmission electron microscope (TEM, JEOL, JEM-1010) was employed to confirm the homogeneous dispersion of MnO_2 nanoparticles on the GO sheets. X-ray photoelectron spectroscopy (XPS, Al Kα, Thermo Scientific) was used to analyze the chemical states of the MnG samples. X-ray diffraction (XRD, Rigaku SmartLab) with Cu Kα radiation was conducted to examine variations in crystallinity. Raman Spectroscopy was performed to investigate the chemical interactions within the MnG samples. Electron paramagnetic resonance spectroscopy (EPR) was utilized to confirm the presence of unpaired electrons.

2.4. Electrochemical measurements

The MnG samples were tested as anode material for 2032-type lithium-ion batteries (LIBs). To prepare the anode, a slurry was prepared by mixing the active material, carbon black (super P), and polyvinylidene fluoride (PVDF) with a weight ratio of 8:1:1 in *N*-methyl pyrrolidone (NMP) solvent. The slurry was then coated onto copper foil and dried at 80 °C under vacuum conditions overnight. The electrode was punched to a diameter of 15 mm, and the loading of the active material was approximately 1.0 mg cm^{-2}. The electrolyte was composed of 1 M $LiPF_6$ dissolved in a mixture of ethylene carbonate and diethyl carbonate, a volume ratio of 1:1. Coin cells were assembled using Li metal foil as the counter electrode, Celgard 2400 polypropylene as the separator, and the electrolyte in an argon-filled glovebox. Galvanostatic charge-discharge and cyclic voltammetry (CV) tests were performed using an automatic battery testing system (WBSC 3000 L, Wo A Tech) in the voltage range of 0.01–3.0 V (vs. Li/Li^+). Galvanostatic tests were conducted at current densities ranging from 0.1 to 2 A g^{-1}, and CV tests were performed at a scan rate of 0.2 mV s^{-1}. Furthermore, the lithium-ion diffusion coefficient was evaluated using the galvanostatic intermittent titration technique (GITT) by applying a series of current pulses corresponding to a voltage change of 10^{-3} V, followed by a 10 min relaxation period between each pulse. Electrochemical impedance spectroscopy (EIS) was performed on an electrochemical workstation (PGSTAT302N, Metrohm Autolab) in the frequency range of 0.01 Hz to 1 MHz with an AC amplitude of 5 mV.

2.5. Computational method

Density functional theory (DFT) calculations were performed using Vienna Ab initio Simulation Package (VASP) [44]. The generalized gradient approximation (GGA) with the Perdew–Burke–Ernzerhof (PBE) functional was used to describe electron exchange-correlation effects [45]. To properly describe the localized nature of Mn 3d states, a Hubbard U correction of U_{eff} = 3.9 eV was applied to Mn [46]. The interaction between core and valence electrons was treated using the projector-augmented wave (PAW) method, with a plane-wave basis set and an energy cutoff of 520 eV [47]. The convergence criterion for electronic calculations was set to 10^{-5} eV with a Hellmann–Feynman atomic force tolerance of 0.01 eV Å^{-1} and a k-point grid of $3 \times 3 \times 8$ was used [48]. Climbing image nudged elastic band (CI-NEB) calculations were performed to investigate Li migration within the α-MnO_2 tunnels [49]. A $1 \times 1 \times 5$ supercell was generated from the α-MnO_2 unit cell ($9.72 \times 9.72 \times 2.85$ Å). The initial Li position was set at the most energetically favorable 8 h site within the α-MnO_2 tunnel [50,51]. Three intermediate images were used for pristine α-MnO_2, whereas seven intermediate images were used for α-MnO_2 containing an O vacancy. For the CI-NEB calculations, an atomic force tolerance of 0.01 eV Å^{-1} and a k-point grid of $2 \times 2 \times 1$ were used.

3. Results and discussion

3.1. Aging-driven structural evolution and defect chemistry toward electrochemically active oxygen-vacancy-rich α-MnO_2

The synthesis procedure of aging-induced MnG samples is schematically illustrated in Fig. 1(a). MnG samples were synthesized using a modified Hummers' method for graphene oxide (GO), in which the conventional H_2O_2 reduction step was intentionally omitted. In the standard procedure, hydrogen peroxide removes residual $KMnO_4$ and reduces MnO_2 byproducts to obtain pristine GO sheets [39]. By excluding this step, MnO_2 nanoparticles were deliberately preserved, enabling the direct formation of MnO_2/GO composites.

An additional aging step was introduced to regulate the structural evolution of manganese oxides and to control defect formation. Immediately after $KMnO_4$ addition, the highly oxidative environment favors the formation of manganese–oxygen species with relatively low defect concentrations, which typically exhibit limited electrochemical activity for lithium storage. Without sufficient aging, these species remain dominant, leading to inferior electrochemical performance, as observed for MnG_0h. Extended aging promotes continuous secondary reactions among residual oxidants, water, and intermediate manganese species, gradually driving the system toward the formation of electrochemically active α-MnO_2. This structural evolution is supported by HR-TEM images in Fig. 1b and c. The nucleation density of MnO_2 nanoparticles increases significantly as the aging time is extended to 72 h, while maintaining a uniform dispersion on GO sheets. Lattice fringes with a spacing of 0.310 nm, corresponding to the (310) plane of α-MnO_2, are consistently observed (Fig. S1), confirming the formation of α-MnO_2 in the aged samples [52,53].

The progressive formation of α-MnO_2 is further corroborated by XRD analysis (Fig. 1d), which displayed characteristic diffraction peaks corresponding to crystalline α-MnO_2 in all aged samples [54]. Scherrer analysis (Fig. S2) revealed gradual crystallite growth from 6.5 nm for MnG_0h to 10.8 nm for MnG_72h, indicating structural maturation during the aging process. FE-SEM images (Fig. S3) and EDS mapping (Fig. S4) further confirmed a steady increase in Mn content from 18.1 at.% to 29.2 at.% with prolonged aging. Collectively, these results demonstrated that the aging process acts as a critical design parameter, transforming initially formed manganese–oxygen species into structurally evolved and electrochemically active α-MnO_2 nanoparticles, thereby establishing a foundation for subsequent defect engineering and performance enhancement [55].

Beyond phase evolution, prolonged aging exerted a profound influence on the defect chemistry of α-MnO_2. Electron paramagnetic resonance (EPR) spectroscopy revealed a characteristic signal at a g-factor of

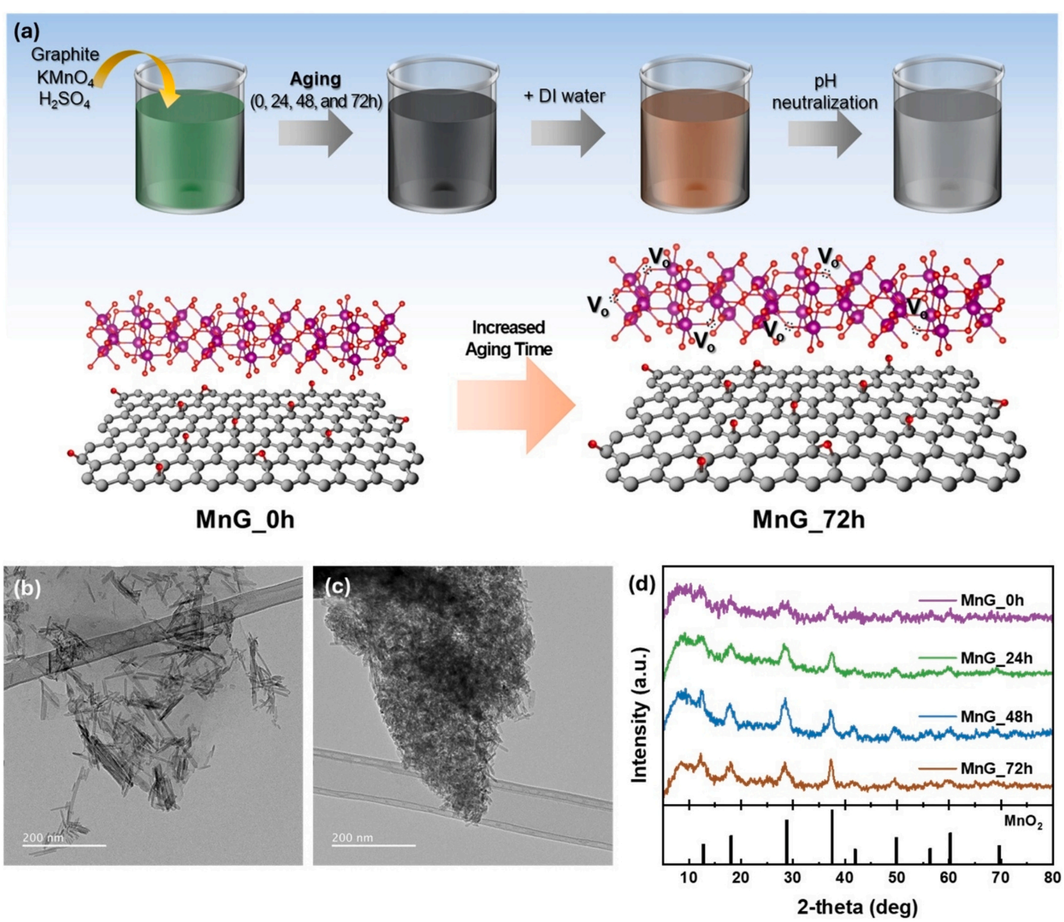

Fig. 1. (a) Schematic illustration of the synthesis procedure of MnG samples. (b, c) HR-TEM images of (b) MnG_0h and (c) MnG_72h samples. (d) XRD spectra of MnG_0h, MnG_24h, MnG_48h, and MnG_72h.

2.003 for all samples (Fig. 2a), which is commonly associated with oxygen-vacancy-related paramagnetic centers [56,57]. Notably, the signal intensity increased with aging time, indicating an enrichment of vacancy-type defects. To further elucidate the origin of these defects, X-ray photoelectron spectroscopy (XPS) was employed to examine the chemical states of Mn species. The O 1 s spectra (Fig. 2b) were deconvoluted into contributions from Mn–O–Mn lattice oxygen, Mn–O–C interfacial oxygen, and Mn–O–H species associated with oxygen-deficient environments [58–60]. Quantitative analysis (Fig. S5) revealed a pronounced decrease in the Mn–O–Mn fraction from 39.3% in MnG_0h to 19.7% in MnG_72h, accompanied by a substantial increase in Mn–O–H from 44.3% to 62.9%, suggesting progressive lattice oxygen depletion in MnO_2. Consistently, the Mn 2p spectra (Fig. 2c) showed a gradual increase in the Mn^{3+}/Mn^{4+} ratio, reaching 1.56 for MnG_72h, while the Mn 3 s-derived average oxidation state significantly decreased from 3.42 to 3.15 (Fig. 2d) [61]. The reduction of Mn species provides charge compensation for oxygen loss, which is characteristic of oxygen vacancy formation in MnO_2 lattices.

The strong correlation between the evolution of Mn oxidation states and the increasing EPR intensity indicates that vacancy formation within the MnO_2 lattice plays a major role in the observed defect enrichment. Significantly, this vacancy-rich state was achieved through a facile aging process without the need for high-temperature reduction or aggressive chemical treatments. Such a defect-engineered structure is expected to provide enhanced electronic conductivity and accelerated lithium-ion transport pathways essential for achieving near-theoretical storage capacity.

3.2. Aging-driven oxygen vacancies enable near-theoretical lithium storage

The electrochemical implications of aging-induced oxygen vacancies were systematically evaluated using galvanostatic charge–discharge and rate capability measurements (Fig. 3a and b). The discharge capacity of MnG_xh increased progressively with aging time, ultimately reaching 1175 mAh g^{-1} at 0.1 A g^{-1} for MnG_72h. This value corresponds to 95.4% of the theoretical capacity of MnO_2 (1232 mAh g^{-1}), suggesting that oxygen-vacancy defects enable more efficient utilization of the MnO_2 lattice through enhanced reaction kinetics and increased accessibility of redox-active sites. Although aging led to a moderate increase in MnO_2 content (Figs. S2-S4), the capacity enhancement cannot be solely attributed to increased MnO_2 loading or particle growth. Instead, the results point to the generation of additional redox-active sites, which are introduced through the formation of oxygen vacancies. This interpretation is consistent with the progressive increase in vacancy-related EPR signals and the Mn^{3+} enrichment observed in XPS analyses (Fig. 2a–d), which collectively indicate the activation of reversible Mn^{3+}/Mn^{4+} redox reactions within the MnO_2 framework. This interpretation is further supported by scan-rate-dependent CV analysis (Figs. S6 and S7), in which the b-values for the oxidation peaks (peaks 2 and 3) increased from 0.49 and 0.58 for MnG_0h to a uniform 0.66 for MnG_72h. The higher b-values signify an increased contribution of surface-controlled pseudocapacitive charge storage, suggesting that oxygen-vacancy engineering facilitates the accessibility of electrochemically active Mn sites and promotes surface-dominated Li-storage kinetics [62,63].

Rate capability measurements further elucidated the critical role of oxygen vacancies in improving electrochemical kinetics. MnG_72h exhibited a high discharge capacity of 425 mAh g^{-1} even at a high

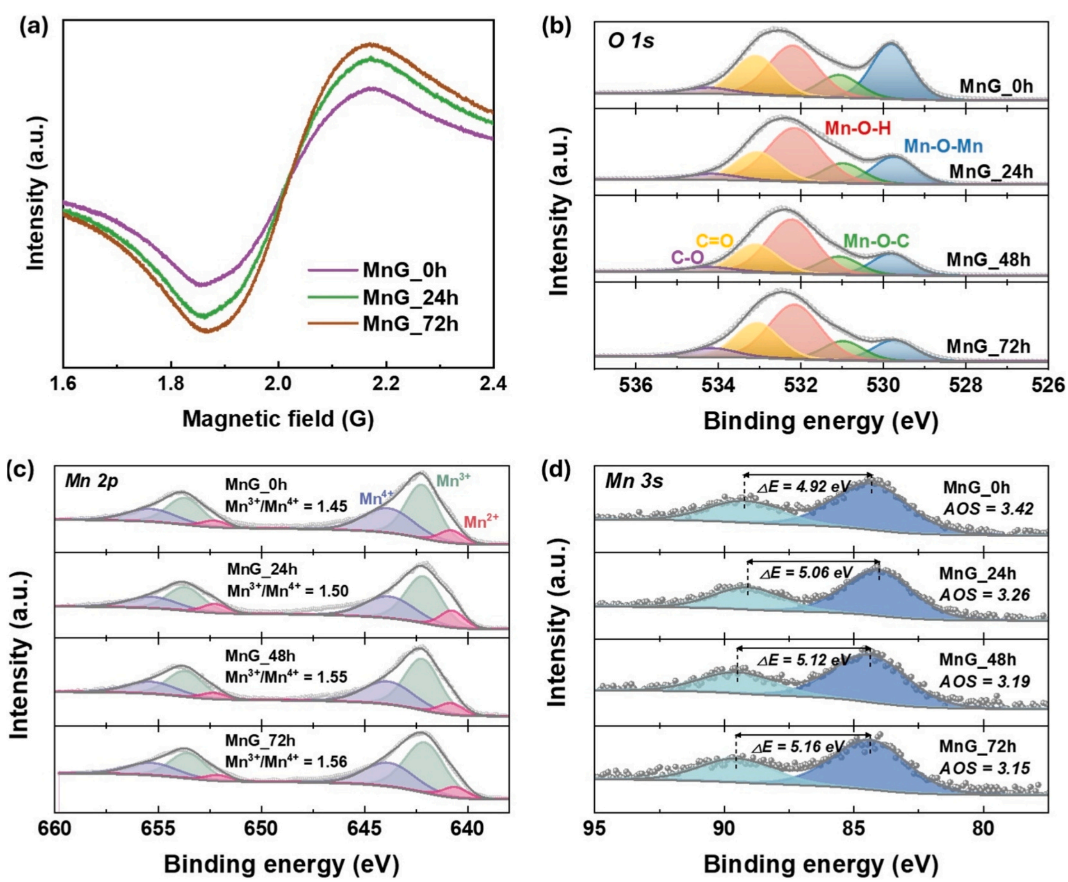

Fig. 2. (a) EPR spectra of MnG samples depending on the aging time. (b-d) XPS spectra: (b) O 1 s, (c) Mn 2p, and (d) Mn 3 s of MnG_0h, MnG_24h, MnG_48h, and MnG_72h.

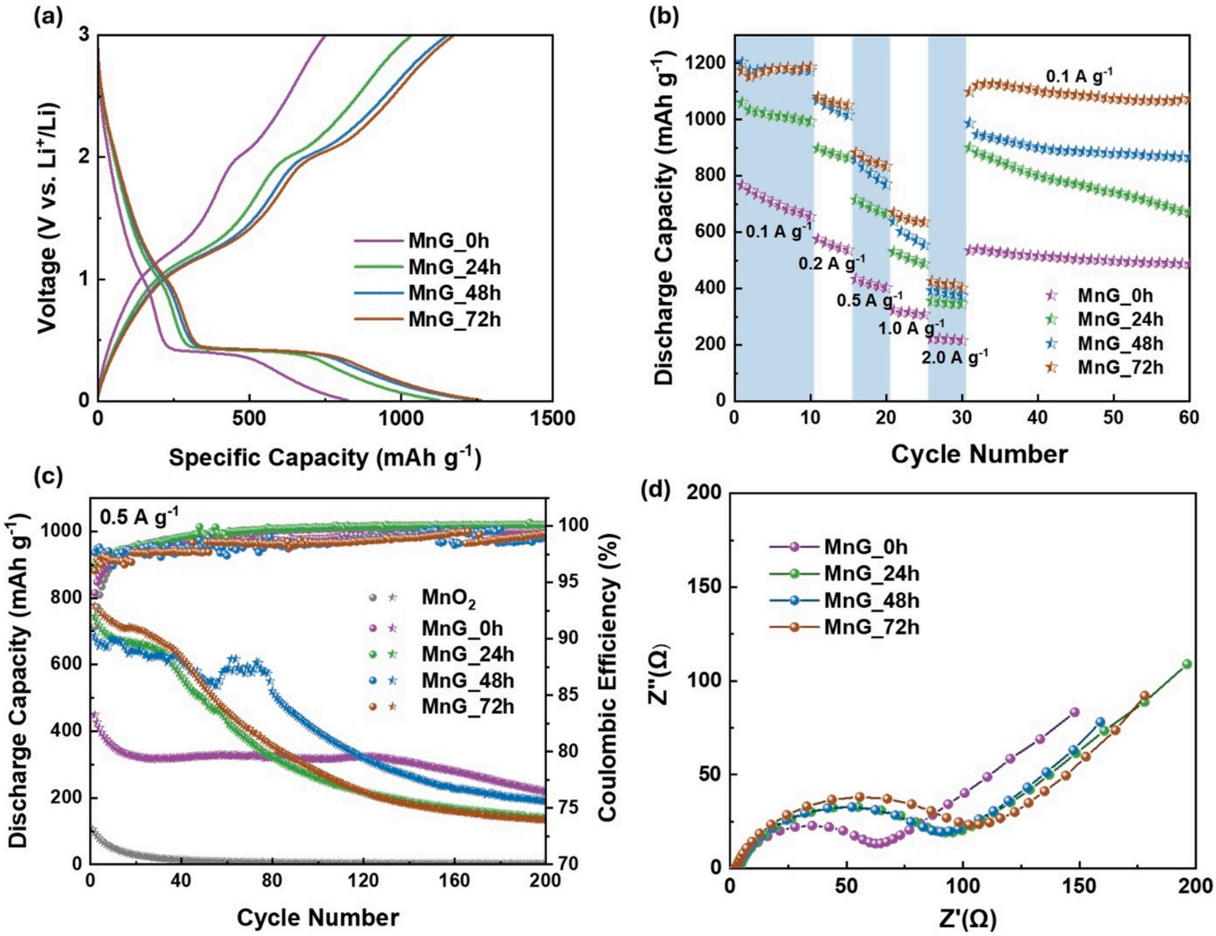

Fig. 3. (a) Galvanostatic charge–discharge curves, (b) rate capability, (c) cyclic stability and coulombic efficiency at 0.5 A g^{-1}, and (d) Nyquist plots measured at open-circuit voltage for MnG samples with different aging times (MnG_xh).

current density of 2 A g^{-1}, outperforming the other MnG_xh samples. In contrast, MnG_0h exhibited severe capacity decay under identical conditions, reflecting sluggish charge transport and limited ion accessibility in vacancy-poor MnO_2. To evaluate the long-term cycling stability, cycle-life tests were conducted at 0.5 A g^{-1} using a galvanostatic charge-discharge method. All MnG samples exhibited coulombic efficiencies approaching 100% during cycling, indicating high electrochemical reversibility. However, MnG samples aged for longer (24, 48, and 72 h) showed relatively low discharge capacities after 200 cycles with capacity retention of ~28%, whereas MnG_0h maintained a higher retention of 48.7%. These results suggest that the additional MnO_2 nanoparticles generated during prolonged aging suffer from limited structural stability during repeated lithiation and delithiation, even though beneficial for achieving high initial capacity. Pronounced capacity fading between 40 and 60 cycles further indicated the instability of the defect-rich MnO_2 formed during the aging process.

Electrochemical impedance spectroscopy (EIS) provided additional insight into the charge-transport characteristics (Fig. 3d). MnG_72h showed a slightly higher charge-transfer resistance (R_{ct}) than MnG_0h. Although oxygen vacancies are expected to improve intrinsic electronic conductivity, the increased surface coverage of MnO_2 nanoparticles may partially hinder direct electron transport through the graphene network, resulting in a slightly higher apparent R_{ct}.

To elucidate the role of oxygen vacancies in governing the transport properties of MnO_2, DFT calculations were performed to examine both electronic and ionic transport characteristics. Fig. 4(a) and (b) show the density of states (DOS) of pristine α-MnO_2 and α-MnO_2 containing an oxygen vacancy, respectively, to examine their electronic properties. Pristine MnO_2 exhibits intrinsic spin polarization originating from high-spin Mn^{4+} ions. Upon the introduction of oxygen vacancies, the electronic structure is substantially altered, as reflected by a pronounced reduction of the bandgap from 1.42 eV to 0.36 eV and the emergence of vacancy-induced electronic states in the vicinity of the Fermi level. These changes indicate vacancy-driven electronic reconstruction within the MnO_6 framework, leading to a redistribution of spin-resolved density of states near the Fermi level and suggesting enhanced electronic transport under oxygen-deficient conditions.

In parallel, the Li-ion transport properties within the MnO_2 framework were evaluated using the CI-NEB method. As shown in Fig. 4(c) and (d), the introduction of an O vacancy along the diffusion pathway reduces the Li migration barrier from 0.17 eV to 0.14 eV. As illustrated in Fig. 4(e) and (f), this reduction is attributed to a local modification of the electrostatic environment along the migration path, whereby the diffusing Li ion experiences weakened repulsive interactions with surrounding oxygen atoms, resulting in a lower energetic penalty at the transition state. Collectively, these results demonstrate that oxygen vacancies play a dual role in MnO_2 by simultaneously enhancing electronic transport through bandgap narrowing and facilitating Li-ion diffusion via reduced migration barriers, thereby improving the overall transport properties of the MnO_2 framework.

3.3. *Low-temperature heat treatment activates MnO_2/graphene interfaces without sacrificing oxygen vacancies*

While prolonged aging established a defect-rich α-MnO_2 framework that enabled high capacity, its limited structural stability during cycling

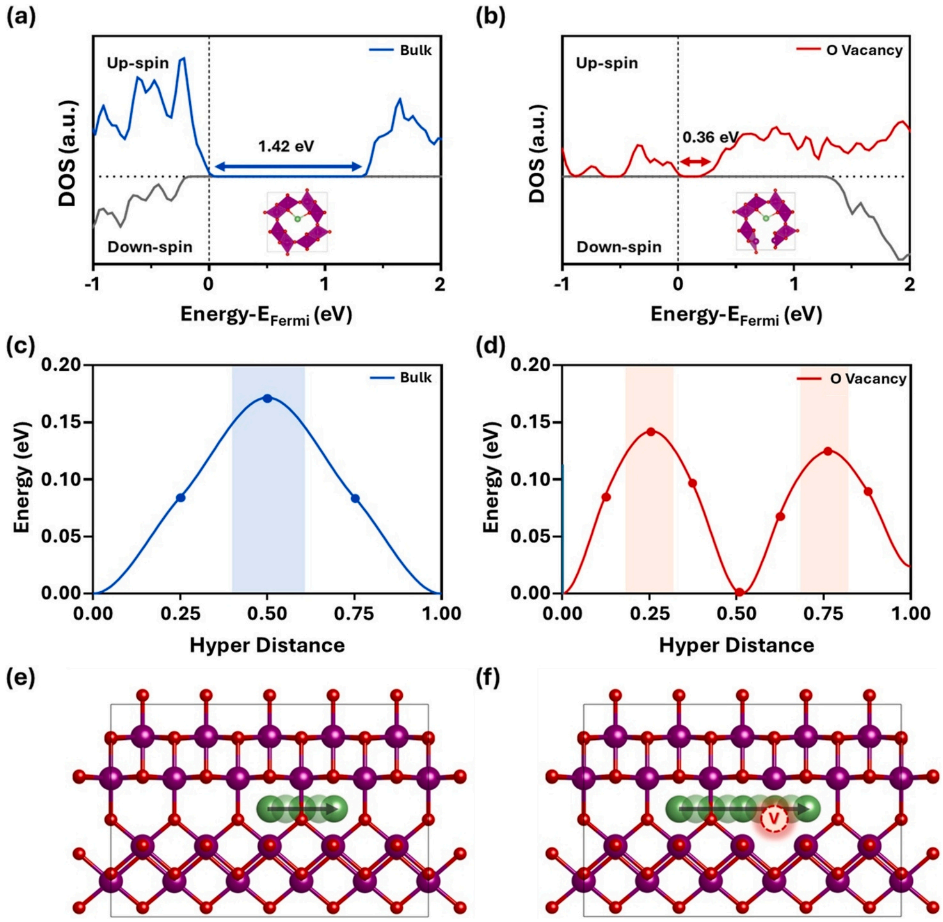

Fig. 4. Electronic structure and Li-ion migration behavior of pristine α-MnO_2 and α-MnO_2 containing an oxygen vacancy. (a,b) Density of states (DOS), (c,d) Li-ion migration energy profiles, and (e,f) Li-ion migration pathways. Panels (a), (c), and (e) correspond to pristine α-MnO_2, whereas panels (b), (d), and (f) correspond to α-MnO_2 containing an oxygen vacancy.

necessitated further interfacial engineering. In particular, weak interfacial coupling between MnO_2 nanoparticles and the graphene framework may contribute to structural degradation during repeated cycling. Therefore, a mild low-temperature heat treatment was introduced to strengthen the MnO_2/graphene interface while preserving the vacancy-rich structure. To decouple aging and thermal effects, MnG_72h was subjected to mild heating at 95 °C for 3 h, yielding MnG_72h_95°C_3h. This treatment primarily enhanced interfacial coupling between MnO_2 nanoparticles and partially reduced graphene sheets.

TEM and XRD analyses (Figs. S8 and S9) confirmed that the low-temperature treatment didn't alter particle size, crystallographic facets, or overall morphology. Instead, subtle interfacial modifications were introduced. Nitrogen adsorption–desorption isotherms and BJH pore size distribution analysis (Fig. S10) further revealed a substantial increase in the BET specific surface area from 36.7 to 187.2 m^2 g^{-1} after the mild heat treatment, accompanied by a significant increase in mesopore volume. Both MnG_72h and MnG_72h_95°C_3h maintained characteristic type IV isotherms, confirming the preservation of the mesoporous framework. This surface area enhancement is primarily attributed to the desorption of residual moisture and weakly adsorbed hydroxyl species retained from the modified Hummers' process, which effectively opens previously clogged pore channels. Simultaneously, interfacial restructuring driven by Mn–O–C bond formation prevents localized sheet aggregation and improves active-site accessibility without altering the bulk crystal structure. Raman spectra (Fig. 5a) showed a decreased I_D/I_G ratio after heat treatment, indicating partial reduction of GO. Both MnG_72h and MnG_72h_95°C_3h exhibited characteristic α-MnO_2 peaks, including the predominant symmetric stretching vibration of Mn—O in MnO_6 octahedra at 643 cm^{-1}. These peaks exhibited a slight negative shift compared with bulk MnO_2, consistent with the presence of oxygen vacancies. Additional bands at 302 and 357 cm^{-1} corresponded to Mn—O lattice vibrations and Mn–O–Mn bending modes, respectively [64]. XPS O 1 s spectra of MnG_72h_95°C_3h (Fig. 5b) confirmed the preservation of oxygen vacancies after heat treatment. MnG_72h_95°C_3h retained a high Mn–O–H fraction (50.5%), while the Mn–O–C contribution increased to 21.8% (compared to 17.4% for MnG_72h), indicating enhanced chemical bonding between MnO_2 and the GO sheet. Mn 2p and Mn 3 s spectra (Fig. S11) further supported the retention of oxygen vacancies, exhibiting a Mn^{3+}/Mn^{4+} ratio of 1.81 and an average oxidation state of 3.10, compared with 1.56 and 3.15, respectively, for MnG_72h.

These interfacial refinements resulted in notable improvements in electrochemical stability without sacrificing capacity. As shown in Fig. 5c, MnG_72h_95°C_3h delivered an initial discharge capacity of 1036 mAh g^{-1} at 0.1 A g^{-1} and exhibited gradual capacity activation during the first 10 cycles. Even at a high current density of 2.0 A g^{-1}, MnG_72h_95°C_3h delivered 505 mAh g^{-1}, outperforming MnG_72h (425 mAh g^{-1}). Long-term cycling at 0.5 A g^{-1} (Fig. 5d) further underscored the stabilizing effect of interfacial engineering. Pristine MnO_2 rapidly degraded from 104 mAh g^{-1} in the first cycle to 2.3 mAh g^{-1} after 160 cycles. MnG_72h showed an initial capacity of 798 mAh g^{-1} but retained only 19.9% of its initial capacity after 160 cycles. In contrast, MnG_72h_95°C_3h exhibited an initial capacity of 931 mAh g^{-1} and underwent sustained activation behavior, reaching 971 mAh g^{-1} at 160 cycles and 977 mAh g^{-1} after 400 cycles. The gradual capacity increase during long-term cycling is commonly observed in conversion-

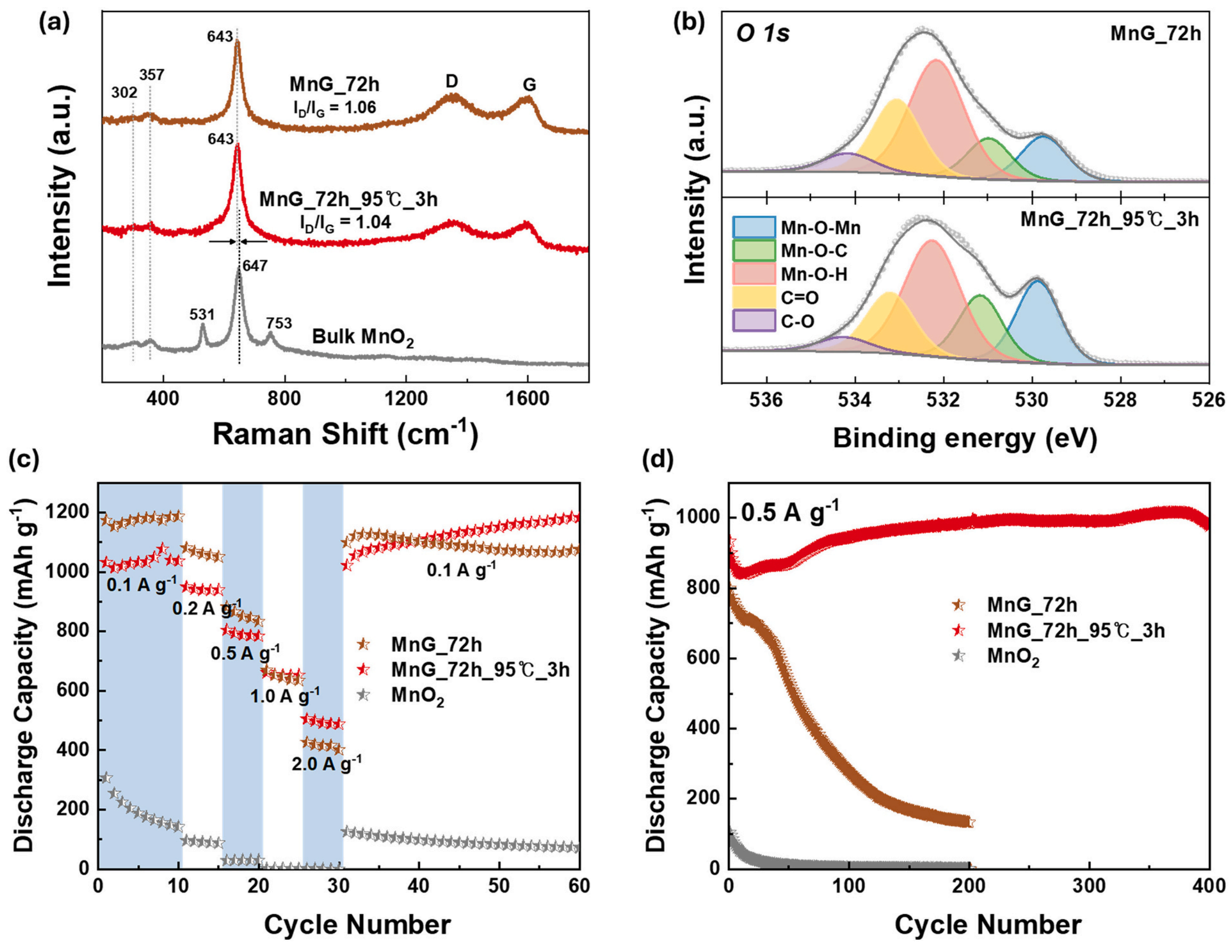

Fig. 5. (a) Raman spectra of MnO_2, MnG_72h, and MnG_72h_95°C_3h. (b) XPS O 1 s spectra of MnG_72h and MnG_72h_95°C_3h. (c) Rate capability of MnO_2, MnG_72h, and MnG_72h_95 °C_3h electrodes measured at current densities of 0.1, 0.2, 0.5, 1.0, and 2.0 A g^{-1}. (d) Long-term cycling performance of MnO_2, MnG_72h, and MnG_72h_95 °C_3h electrodes measured at a constant current density of 0.5 A g^{-1}.

type anodes and is mainly attributed to the progressive reactivation of active materials and additional lithium storage at newly generated interfaces during repeated cycling [65–67]. While pristine MnO_2 electrodes are known to suffer from severe volume expansion (~200%) during lithiation, in the post-mortem FE-SEM images after 120 cycles (Fig. S12), MnG_72h and MnG_72h_95°C_3h exhibit markedly reduced volumetric expansion, with estimated expansion ratios of 163% and 148%, respectively. This reduction can be primarily attributed to strong Mn–O–C interfacial bonding, which promotes more uniform lithiation and suppresses localized strain. The outstanding cycling durability of MnG_72h_95°C_3h was further evaluated by comparison with recently reported MnO_2-based anode materials (Table S1). Notably, MnG_72h_95°C_3h maintained a high reversible capacity of 977 mAh g^{-1} after 400 cycles at 0.5 A g^{-1}, demonstrating remarkable long-term cycling durability compared with previously reported MnO_2-based anodes.

Enhanced electrochemical kinetics are further supported by EIS and GITT analyses (Fig. S13 and S14). Notably, both samples exhibit nearly identical series resistance (R_s), indicating that the kinetic improvements originate from intrinsic structural evolution of the electrode rather than differences in external contact resistance. MnG_72h_95°C_3h exhibits a significantly lower R_{ct} (80 Ω) compared to MnG_72h, as evidenced by the markedly reduced diameter of the high-frequency semicircle. This suggests that mild thermal treatment effectively optimized the internal interfacial contact between the manganese oxide and the carbon matrix, facilitating rapid charge exchange. Consistent with the EIS results, GITT analysis reveals that MnG_72h_95°C_3h maintains a superior lithium-ion diffusion coefficient (D_{Li+}) across the entire voltage range. The synergistic effects of the enriched oxygen vacancies and the strengthened MnO_2/graphene interface promote rapid charge transfer and facilitate Li-ion transport within the electrode. Consequently, these improved kinetic parameters provide a robust explanation for the performance of our sample in achieving near-theoretical capacity and exceptional rate capability.

4. Conclusion

In summary, we demonstrated an aging-driven defect engineering approach for the controlled synthesis of oxygen-vacancy-rich α-MnO_2/graphene composites that exhibited high lithium-storage capacity, fast electrochemical kinetics, and robust cycling stability. Systematic variation of the aging time in a modified Hummers' process revealed that aging governed the structural evolution of manganese oxides by regulating both phase development and defect formation, rather than serving as a passive processing step.

Prolonged aging promoted continuous secondary reactions that favored the formation of α-MnO_2 nanoparticles with abundant oxygen vacancies, resulting in improved intrinsic electronic conductivity and facilitated lithium-ion transport. As a result, the defect-rich composites delivered a near-theoretical reversible capacity of 1175 mAh g^{-1} and retained excellent rate capability at high current densities. Furthermore, a subsequent low-temperature heat treatment selectively activated the MnO_2/graphene interface while preserving oxygen vacancies, leading to outstanding long-term cycling stability.

CRediT authorship contribution statement

Seon-Yeong Lee: Writing – original draft, Visualization, Validation, Investigation, Formal analysis. **Myung kyoon Kim:** Writing – original draft, Validation, Investigation, Formal analysis. **Jonghun Shin:** Writing – original draft, Visualization, Software. **Giwon Lee:** Validation, Formal analysis. **Hae-ri Lee:** Validation, Methodology. **Juyeon Hong:** Validation. **Hyeokjun Yoo:** Validation. **Seunggyun Han:** Formal analysis. **Uhyeok Son:** Formal analysis. **Haesun Park:** Writing – review & editing, Validation, Supervision, Resources. **Han-Ik Joh:** Writing – review & editing, Validation, Supervision, Resources, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.est.2026.124426.

Data availability

Data will be made available on request.

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